Arista Access Node SOP for Critical Optical Transport Failure MALL-ACC-21

Node: MALL-ACC-21 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_6662

Description

This Standard Operating Procedure (SOP) outlines the steps to resolve critical optical transport failures on the Arista Access Node MALL-ACC-21, which is part of the Malleswaram Node in the Telecom network. The failure domain in question is Optical Transport Failure, specifically related to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. This is a critical failure that requires immediate attention to prevent network outages and data loss. The Arista Access Node, being a critical component in the access network, relies on optical transport for connectivity to other nodes. Any failure in this domain can cause a cascading effect, leading to network instability and downtime. The root cause of this failure is often attributed to hardware issues, such as faulty SFP/XFP modules or fiber cuts, which can be exacerbated by environmental factors like temperature fluctuations, humidity, or physical damage. To diagnose and resolve this issue, it is essential to understand the underlying protocol mechanics and hardware components involved. The Arista Access Node uses IEEE 802.1Q and IEEE 802.3ad protocols for link aggregation and trunking, respectively. The SFP/XFP modules used in this node are designed to operate within specific power and temperature ranges. Any deviation from these ranges can cause the module to malfunction or fail. In this SOP, we will outline the steps to diagnose and resolve the critical optical transport failure on the Arista Access Node MALL-ACC-21.

Pre-requisite Checks

- Verify that the Malleswaram Node is accessible and that the dispatch team has been notified.

- Check the redundant link status to ensure that there are no other issues in the network.

- Verify that the Arista Access Node MALL-ACC-21 is powered on and that the fans and power supplies are functioning properly.

Potential Root Causes

Cause	Probability
Faulty SFP/XFP module	High
Fiber cut or damage	Medium
Rx/Tx power loss due to environmental factors	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware sfp detail, Expected Output: SFP details, including temperature, voltage, and transmit/receive power levels, Description: This command displays detailed information about the SFP/XFP modules, including temperature, voltage, and transmit/receive power levels. This information can help identify any issues with the
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modules, such as overheating or power loss.

Command: show mlag brief, Expected Output: MLAG configuration and
status, Description: This command displays the MLAG configuration and
status, including the peer device and link status. This information
can help identify any issues with the link aggregation protocol.

Command: show interfaces, Expected Output: Interface configuration and
status, Description: This command displays the interface configuration
and status, including the speed, duplex, and link status. This
information can help identify any issues with the network interfaces.
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Resolution Steps

Step 1: Step: Replace faulty SFP/XFP module, Command: replace hardware sfp , Description: Replace the faulty SFP/XFP module with a new one. Ensure that the new module is of the correct type and is properly seated in the module slot.

Step 2: Step: Verify fiber integrity, Command: show interfaces, Description: Verify that the fiber is intact and that there are no issues with the link. Check the link status and speed/duplex settings.

Step 3: Step: Adjust Rx/Tx power levels, Command: config t interface channel-group speed delay restore exit save reload, Description: Adjust the Rx/Tx power levels to optimal settings. This may involve adjusting the channel-group settings, speed, and delay restore settings.

Post-Check Verification

```
Command: show hardware sfp detail, Expected Output: SFP details,
including temperature, voltage, and transmit/receive power levels,
Description: Verify that the SFP/XFP module is functioning properly
and that the temperature, voltage, and transmit/receive power levels
are within the recommended range.

Command: show mlag brief, Expected Output: MLAG configuration and
status, Description: Verify that the MLAG configuration and status are
correct, including the peer device and link status.

Command: show interfaces, Expected Output: Interface configuration and
status, Description: Verify that the interface configuration and
status are correct, including the speed, duplex, and link status.
```

Roll-back Procedures

- Step: Restore original SFP/XFP module, Command: replace hardware sfp , Description: Restore the original SFP/XFP module to its original position.

- Step: Revert interface configuration, Command: config t interface channel-group speed delay restore exit save reload, Description: Revert the interface configuration to its original settings, including the channel-group settings, speed, and delay restore settings.